

VINAY VISHNU PRABHU

Senior Backend & Platform Engineer | Python | Distributed Systems | AWS | Cloud | AI Engineering

Goa, India | Fully remote since 2021 | UTC+5:30, overlaps to 12pm ET | +91 70202 01272

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SUMMARY

Senior Backend & Platform Engineer with 5+ years building distributed systems and cloud infrastructure on AWS across IoT, SaaS and EV platforms - currently running the battery-swap and telemetry platform behind 2,000+ vehicles and 180+ stations in East Africa. I find and fix the problems teams have learned to live with: a fraud detection system that surfaced a revenue leak nobody could see, a cost flaw caught before build, a file-descriptor leak that had been restarting production every week. Promoted three times in under 5 years, intern to engineering lead; led a 5-engineer team and ran 70+ technical interviews.

EXPERIENCE

Zeno (EV Infrastructure) | Senior Software Engineer | Remote | Dec 2025 - Present

IoT battery-swap and fleet-telemetry platform across East Africa - 2,000+ electric bikes, 180+ swap stations, 1,200+ swaps/day.

- **Built the platform's first swap-fraud detection system, surfacing a revenue leak that had been invisible** - an extensible rule engine correlating 4-5 telemetry signals per transaction, designed from production data with no labelled fraud set or prior baseline. In its first two weeks it screened 33,000+ swaps, confirmed 30+ fraud cases and flagged 600+ for review, giving operations the first quantified view of fleet-wide fraud. Automated warning notifications are the next layer in build.
- **Cut a core production service from 40% average and 65% peak CPU to 1.1% and 6.7%**, freeing co-located services that had been starved of CPU and unable to execute reliably. Built a pidstat watcher sampling across three separate days and time windows to identify the real offenders first, moving the team from assumption-based optimisation targets to data-driven ones.
- **Avoided a projected 80-85% higher cloud run-rate** by catching a cost flaw in a planned Partner API architecture before build and re-architecting it as a dual-mode pull/push service in FastAPI (AWS EC2, Azure Event Hubs, retry caching, lookup-table optimisation). Raised the problem unprompted and delivered the replacement design in 3 days against a 3-week estimate.
- **Introduced the organisation's first CI/CD platform** on GitHub Actions with automated rollback, replacing manual releases with zero-touch deploys across Dev/QA/Prod - recovering 1-2 engineer-days per release cycle for feature work and making every-sprint releases routine. Authored the runbooks and ran the knowledge-transfer sessions that drove team-wide adoption.
- **Replaced LiteFS Consul-based leasing** with a custom lightweight leasing implementation in Go, removing Consul entirely from the deployment topology and cutting both memory overhead and lease-acquisition latency across multi-region SQLite replication clusters - shipped in-house rather than waiting on an upstream fix.
- **Eliminated 3-4 emergency production restarts per week, with zero recurrences since**, by root-causing a file-descriptor leak across the full code path - accelerated by an internal AI code-analysis agent I built - and shipping the fix overnight.
- **Secured application access for ~200 users across 4 enterprise partners** by implementing unified Microsoft and Google SSO behind a single authentication endpoint with granular backend RBAC, closing legacy security vulnerabilities.

Infuse (useMango) | Goa, India | Remote | Feb 2021 - Oct 2025 | 4 yrs 9 mos

Promoted three times: Intern to Associate to Software Engineer to Senior Engineer & Lead.

Senior Software Engineer & Lead, Infuse | Jul 2024 - Oct 2025

- **Increased platform throughput 40% and cut execution failure rates** on UMAI, a Python low-code test automation platform processing thousands of daily executions, by decomposing the backend into Python microservices and event-driven pipelines on AWS DynamoDB and SQS.
- **Led a 3-engineer team to 90% of MVP scope across 5 consecutive sprints** under aggressive timelines through lean planning and tight scope control - owning the engineering standards the team ran on plus 1:1s, performance reviews and growth plans.
- **Shipped two high-priority AI proofs of concept - self-driving and self-healing tests - 2 months ahead of a 3-month deadline**, establishing a rapid-prototyping methodology later adopted into the core product roadmap.

- **Eliminated environment drift across CI/CD and production** by codifying infrastructure with AWS CDK and standardising a containerised Linux deployment stack, after auditing and re-estimating 18+ epics to correct over-optimistic architecture assumptions that were blocking reliable delivery.
- **Conducted 70+ technical interviews** across the organisation, setting how the team evaluated backend candidates.

Software Engineer, Infuse | Jul 2022 - Jun 2024

- **Reduced manual triage overhead 70%** on a Python/Ruby test automation engine supporting 1000+ daily executions by refactoring monolithic components into containerised microservices and adding real observability.
- **Raised SLA compliance from 30% to 85% within 3 months** while leading a 5-engineer customer support engineering team, through systematic incident response improvements and platform stability work.
- **Reduced batch job completion time 75%**, from 1 hour to under 15 minutes, by re-engineering legacy test execution with asyncio and multi-threaded orchestration.

Associate Software Engineer, Infuse | Jul 2021 - Jun 2022 (joined Feb 2021 as intern, converted to full-time)

Concurrency and performance work on Windows test software in Python, Ruby and C++ - resolved threading deadlocks across 100+ concurrent test cases, cutting runtime failures 40%.

SKILLS

Languages: Python, Go, TypeScript, JavaScript, Ruby, SQL

Backend and APIs: FastAPI, Django, Flask, REST, GraphQL, gRPC, asyncio, WebSockets, microservices, event-driven architecture, Pytest, Playwright

Cloud and DevOps: AWS (EC2, Lambda, S3, SQS, DynamoDB, IAM, Cognito, CloudWatch), Terraform, OpenTofu, AWS CDK, Infrastructure as Code, Docker, Linux, GitHub Actions, Jenkins, CI/CD, Git, Azure Event Hubs

Messaging and Streaming: RabbitMQ, AWS SQS, Azure Event Hubs, message queues, event-driven pipelines

Observability: Grafana, Datadog, CloudWatch, structured logging, incident response, performance profiling

Databases: PostgreSQL, MySQL, DynamoDB, MongoDB (Beanie ODM), Redis, SQLite, LiteFS, Firebase, SQLAlchemy, distributed replication

Architecture and Security: Distributed systems, systems design, platform engineering, caching, OAuth 2.0, SSO, RBAC, JWT

AI Engineering: Anthropic and OpenAI APIs, agent orchestration, tool use, MCP (Model Context Protocol), context engineering

PROJECTS

Jii App - Project Lead and Backend Architect | Beta on Google Play Store and Apple App Store

- Architected and owned end-to-end delivery of a multi-service B2B/B2C SaaS platform for iOS and Android powering e-commerce, short-form video, job listings and real-time chat on FastAPI, AWS and MongoDB. Led a 10-person cross-functional team to 1,000+ beta users at 99% uptime.

Social Agent - Creator and Owner | In production for client work

- Built an agent that produces client social posts end to end, generating copy and imagery together rather than bolting stock images onto text - orchestrating Claude, Higgsfield and Nano Banana behind custom context-driven skills, and running against live client work with real brand-voice, format and failure-mode constraints.

EDUCATION

Master of Computer Applications (MCA) | Goa Business School, Goa University | 2018-2021 | GPA 7.9/10

B.Sc. Electronics | St. Xavier's College, Mapusa, Goa | 2015-2018 | GPA 9.0/10, Distinction

PUBLICATION

"IoT Based Appliance Automation using Raspberry Pi." International Journal of Electronics Engineering, Vol. 10, Issue 2, 2018, pp. 422-425. ISSN 0973-7383.

csjournals.com/IJEE/PDF10-2/55.%20CFP.pdf